Proximal-to-Distal Energy Transfer in the Golf Swing

Where It Occurs, How It Differs with Skill, and When It Should Happen: A Thesis-Length Synthesis with an Executed Counterfactual Modeling Pathway in UpstreamDrift

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Abstract

Proximal-to-distal (P→D) sequencing — the ordered peaking of pelvis, thorax, arm, and club rotational speeds — is the most widely reported organizing principle of the golf downswing, and it is usually read as evidence that mechanical energy should be passed outward along the kinetic chain as early and as completely as possible. This thesis reviews the mechanics and the empirical literature behind that reading and asks the sharper question: *when* during the downswing should the P→D transfer occur? Across double-pendulum mechanics, delayed-release simulation, optimal control, and the segmental kinetic-energy literature, the evidence favors a strategy that *retains* energy in the proximal system early in the downswing and *accelerates* the handoff late. We formalize this retain-early/release-late hypothesis in the language of the zero-torque counterfactual (ZTCF) and zero-velocity counterfactual (ZVCF) decompositions implemented in the UpstreamDrift platform, and we execute a first quantitative test on the platform's backend-neutral two-segment golf model: a wrist-torque timing sweep (92 torque programs), pointwise drift/control counterfactual decompositions, and Robertson-Winter style wrist-interface power accounting, all generated by committed, reproducible scripts. In this model, driving the wrist from the top costs 15% of clubhead speed relative to a fully passive wrist, delaying the wrist drive to the optimal onset gains 12%, and a program that actively restrains the wrist early and drives it late is best of all at both shoulder-torque levels tested. The energy accounting shows the winning programs move *less* energy into the club early and far more late, with the best program performing *negative* wrist work early — active retention. We interpret these results against the literature, state their limitations honestly (a planar 2-DOF model with a fixed hub and rigid shaft), and lay out the validated pathway — re-simulation parity, segmental energy-flow metrics, full-body engines, and motion-capture replication — by which the platform can carry the question to thesis-grade generality. No fabricated data or sources are used anywhere in this document: every reference is verified against a publication record, and every number in the results chapters is produced by code committed alongside this text.

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Provenance and scope

This document is the deliverable of UpstreamDrift epics #8373 and #8383 (thesis expansion with executed modeling pathway). Chapters 1–5 are synthesis: they report the literature and the platform, with every reference verified against a publication record. Chapters 6–8 report *simulation results generated by committed scripts in this repository* (`scripts/research/proximal_distal_energy/`), with parameters and provenance recorded alongside the data. Remaining generalization work is tracked in issues #8377 and #8378. Repository links point to the `main` branch of D-sorganization/UpstreamDrift.

Chapter 1

Introduction

1.1 Motivation

Clubhead speed at impact is the dominant controllable determinant of drive distance, and it correlates strongly with playing standard (Fradkin, Sherman, and Finch 2004; Hellström 2009; Hume, Keogh, and Reid 2005). The golf downswing is among the most-studied examples of a *sequenced* multi-segment rotation in sport biomechanics (McPhee 2022; Dillman and Lange 1994). Across measurement systems and populations, the downswing of a skilled golfer shows a characteristic order: the pelvis reaches its peak angular speed first, then the thorax, then the lead arm, and finally the club, with each successive segment peaking faster than the one before (Cheetham et al. 2008; Tinmark et al. 2010). This pattern — the *kinematic sequence*, an instance of the general proximal-to-distal (P→D) sequencing observed in throwing and striking skills (Putnam 1993) — is routinely interpreted as the visible signature of mechanical energy generated in the large proximal segments and transferred outward to the small, fast distal ones, and it has been translated into coaching doctrine accordingly (Smith et al. 2015).

That interpretation leaves a first-order question unanswered:

> **Should a golfer attempt to transfer energy from proximal to distal segments *early* in the downswing, or should energy be *retained* in the proximal segments early and the transfer deliberately *accelerated* late?**

The two strategies imply opposite coaching cues ("fire the club from the top" versus "hold the lag and let it release"), opposite torque programs at the wrists and torso, and different predictions for what counterfactual decompositions of a measured swing should show. The question also connects to a long-standing thread in the modeling literature: double-pendulum analyses have argued since the 1960s that the highest clubhead speeds arise when the distal joint is *not* driven early — when release is passively or actively delayed (Cochran and Stobbs 1968; T. Jorgensen 1970; Milburn 1982; Pickering and Vickers 1999) — while the coaching translation of the kinematic sequence often emphasizes initiating distal motion promptly after transition.

UpstreamDrift approaches this question with *counterfactual dynamics.* Two decompositions are implemented across its physics engines: the **zero-torque counterfactual (ZTCF)** — the acceleration the system would exhibit at the current instant if all applied joint torques were removed, isolating what accumulated motion, gravity, and inter-segment coupling produce on their own — and the **zero-velocity counterfactual (ZVCF)** — the acceleration the applied torques, together with gravity, would produce at the current pose from rest, removing all motion-dependent effects. Preliminary ZTCF/ZVCF analyses with the platform's MATLAB pipeline motivated the hypothesis examined throughout this document: that *restraining* the P→D handoff early in the downswing and accelerating it late may be the more effective strategy.

1.2 Research questions

The document is organized around four questions:

- **RQ1 — Mechanism.** By what mechanical pathways does energy generated proximally reach the club, and what does each pathway imply about the *timing* of the handoff? (Section 2)
- **RQ2 — Evidence.** Where does the P→D transfer occur in measured swings, and how do skilled and less-skilled players differ — in sequence order, in magnitudes, or in transfer efficiency? (1)
- **RQ3 — Formalization.** How can "transfer early" versus "retain-early/release-late" be stated as falsifiable claims about measurable quantities, and what do the ZTCF/ZVCF counterfactuals predict under each? (Section 4)
- **RQ4 — Demonstration.** What do the platform's own models show when the timing of the distal handoff is manipulated directly, with the energy accounting and counterfactual decompositions computed along every trajectory? (Section 6, Section 7, Section 8)

1.3 Approach and contributions

The document makes six contributions:

1. a synthesis of the mechanics of sequenced swings, from the summation-of-speed principle through segment-interaction dynamics, parametric energy transfer, and optimal control (Section 2);
2. a methodological account of what "energy transfer" means operationally — joint power, joint-force power, segmental energy accounting, induced accelerations — and where the popular kinematic-sequence proxy fails (Section 2.5);
3. a critical review of the empirical evidence, including the directly relevant segmental kinetic-energy sequencing studies (Kenny et al. 2008; Anderson, Wright, and Stefanyshyn 2006) and the skill-level literature (1);
4. a formalization of the timing question and of the ZTCF/ZVCF counterfactual framework, including the distinction between pointwise and re-simulation counterfactuals (Section 4);
5. an **executed modeling pathway**: a reproducible experiment suite on the platform's backend-neutral double-pendulum golf model — 92 torque programs swept over wrist-drive onset time and profile, scored at a consistently defined impact, with pointwise drift/control decompositions and Robertson-Winter wrist-interface power accounting computed along every representative trajectory (Section 6, Section 7); and
6. an honest bridge from these 2-DOF results to full generality: what the simplified model can and cannot establish, and the concrete, already-tracked steps that extend the pathway to re-simulation parity, full-body engines, and motion-capture replication (Section 8, Section 9).

1.4 Reading guide

Readers interested only in the new quantitative material can read Section 4 for the hypothesis, Section 6 for the setup, and Section 7 and Section 8 for the outcome. Readers using this document as a literature resource will find the review chapters (Section 2, 1) and the verified bibliography self-contained. Appendices inventory the platform (4), document reproducibility (**?@sec-appendix-repro**), collect validation evidence (**?@sec-appendix-validation**), and tabulate model parameters (**?@sec-appendix-params**).

Chapter 2

Mechanics of Sequenced Swings

This chapter assembles the mechanical foundations the rest of the document builds on: the dynamics formalism (Section 2.1), the summation-of-speed principle and its dynamical correction (Section 2.2), the double-pendulum tradition that made release timing an explicit variable (Section 2.3), the optimal-control literature (Section 2.4), and the operational meanings of "energy transfer" (Section 2.5).

2.1 Dynamics formalism

Write the equation of motion of an articulated chain in standard manipulator form (Featherstone 2008):

$$M(q)\,\ddot{q} \;=\; \tau \;-\; b(q,\dot{q}), \qquad b(q,\dot{q}) \;=\; C(q,\dot{q})\,\dot{q} \;+\; g(q) \;+\; d(\dot{q}), \tag{2.1}$$

with configuration q, inertia matrix M, applied generalized torques τ, Coriolis/centripetal terms $C\dot{q}$, gravity g, and velocity-dependent dissipation d. Two structural facts drive everything that follows. First, the system is *coupled*: M is not diagonal, so torque at any joint induces acceleration at every joint — the basis of induced-acceleration analysis (Zajac and Gordon 1989) and of the counterfactual decompositions of Section 4. Second, the equation is *affine in* τ: the acceleration splits exactly into a drift part (what accumulated motion and gravity produce alone) and a control part ($M^{-1}\tau$), a property the UpstreamDrift test suite enforces to numerical tolerance (**?@sec-appendix-validation**).

2.2 The summation-of-speed principle and its limits

The oldest formalization of P→D sequencing is the *summation of speed* principle: in skills demanding high end-point speed, each segment should begin its motion at the moment of peak speed of the segment proximal to it, so that distal speed is built on top of proximal speed (Bunn 1972). Milburn applied the principle to golf with a double-pendulum analysis of the downswing and found the characteristic pattern of an initially dominant proximal (arm) rotation followed by rapid distal (club) rotation, noting that *"an initial delay in the uncocking of the wrist"* allows the arm to reach greater acceleration and the club's acceleration to be summed with the proximal segment's maximum (Milburn 1982). Note the structure of that finding: the summation principle, taken naively, prescribes *sequential initiation*; Milburn's analysis showed the benefit comes specifically from *delaying* the distal contribution.

Putnam's segment-interaction analyses put the principle on a dynamical footing and exposed its limits (Putnam 1991, 1993). In a linked chain, each segment's angular acceleration depends not only on joint torques but on *motion-dependent interaction torques* — centripetal and Coriolis terms proportional to squares and products of segment angular velocities (the $C\dot{q}$ term of Equation 2.1). Putnam showed the P→D pattern in striking and throwing arises substantially from these interaction torques: proximal deceleration and distal acceleration late in the motion are coupled through the joint force and occur even without a distal joint torque. Feltner and Dapena's classic analysis of the baseball pitch quantified the same interplay at the shoulder and elbow (Feltner and Dapena 1986), and Herring and Chapman showed in simulated throws that the *timing* of torque onsets and torque reversals — including deliberately reversed (restraining) torques — materially changes end-point speed (Herring and Chapman 1992). Two implications matter for golf. First, kinematic order does not reveal *which* torques produced it: an identical peak sequence can arise from different mixes of muscular and interaction torques. Second, the proximal deceleration observed in experts need not be an active "braking" strategy — it can be the dynamical *consequence* of the distal segment accelerating and back-reacting on the chain.

Marshall and Elliott added a measurement caution: planar peak-ordering analyses miss long-axis rotations, which peak very late in racquet skills and contribute substantially to end-point speed (Marshall and Elliott 2000); the golf analogue is forearm rotation and wrist action in the final phase of the downswing.

2.3 Double-pendulum mechanics of the golf downswing

The two-lever, one-hinge model of the downswing — an "arm" link rotating about a hub, connected by a wrist hinge to a "club" link — has been the workhorse of golf mechanics since Cochran and Stobbs' *Search for the Perfect Swing* (Cochran and Stobbs 1968) and Jorgensen's dynamical analysis fitted to stroboscopic photographs of a professional swing (T. Jorgensen 1970; T. P. Jorgensen 1999); Penner's review surveys the tradition (Penner 2003). Its enduring value here is that it makes the *timing* of the P→D handoff an explicit variable (the "release": the point at which the wrist angle is allowed, or driven, to open) and exposes the mechanism by which delay pays. Three results anchor the analysis:

Delayed release increases clubhead speed. Pickering and Vickers, sweeping release angles in a driven double-pendulum model, found clubhead velocity at impact increases as release is delayed — the classical "late hit" (Pickering and Vickers 1999). Jorgensen's fitted model reproduced the expert swing only with a substantially delayed opening of the wrist angle (T. Jorgensen 1970; T. P. Jorgensen 1999). In these models an early release (early P→D handoff) is strictly inferior: club kinetic energy purchased early is paid for with arm speed that never develops.

Late wrist torque helps; early wrist torque is the wrong tool. Sprigings and Neal, using a torque-driven three-segment model with physiologically bounded actuators, found that an active wrist torque applied *during the latter stages of the downswing* increased clubhead speed by approximately 4% (Sprigings and Neal 2000); Sprigings and MacKenzie confirmed the delayed-release benefit in a follow-up simulation focused on release timing (Sprigings and MacKenzie 2002). The point is not that distal effort is useless, but that its productive window is late.

Energy transfer to the club can be largely passive — governed by the changing geometry. White analyzed an *undriven* double pendulum and showed that the transfer of kinetic energy from arms to club late in the downswing is a parametric-transfer phenomenon: as the club swings out, the effective moment of inertia of the system changes, and conservation of energy and angular momentum move kinetic energy distally without any wrist torque at all; the wrist-cock angle maintained before release is the principal efficiency-determining parameter (White 2006). Miura's analysis of the "inward pull" complements this: an expert's hands moving toward the body near impact — shrinking the hub radius when centrifugal loading is fully developed — performs work against the centrifugal force that appears as additional clubhead kinetic energy (Miura 2001). Nesbit and McGinnis' hub-path analyses generalize the same idea: the non-circular hand path of skilled golfers, including its late-downswing tightening, is a mechanism for shaping kinetic transfer from golfer to club (Nesbit and McGinnis 2009; Nesbit and McGinnis 2014).

Together these results describe a machine whose highest-output mode is: *load the proximal rotation first, hold the distal joint folded (retaining energy in a compact, fast-rotating configuration), then let geometry and — in the last phase — modest distal torque move the energy outward rapidly.* The double-pendulum literature is already an argument for **retain-early / release-late**. Section 7 tests exactly this proposition on the UpstreamDrift implementation of the model.

2.4 Optimal control and torque sequencing

If the timing question is posed as an optimization — what torque program maximizes clubhead speed subject to actuator limits? — the answer has been studied repeatedly. Lampsa's early optimal-control study of a driven double pendulum computed torque histories maximizing drive distance, and notably found the *optimal* torques differed in structure from those measured (via inverse dynamics) in his own swing (Lampsa 1975) — early evidence that human swings need not be optimal and that the optimum is a meaningful target rather than a redescription of expert behavior. Sharp, fitting shoulder-arm-club models to expert swing data and then optimizing torque programs, established a pattern for the best use of the available driving torques across the downswing phases, with the distal contribution structured late rather than distributed evenly (Sharp 2009). Three-dimensional forward-dynamics models extended the approach with anatomically meaningful actuators and flexible shafts: MacKenzie and Sprigings' 3D model (MacKenzie and Sprigings 2009), the carry-optimized model of Balzerson, Banerjee and McPhee (Balzerson, Banerjee, and McPhee 2016), and the six-degree-of-freedom dynamic optimization of McNally and McPhee (McNally and McPhee 2018) all generate expert-like clubhead speeds with torque programs whose distal elements act late in the downswing. McPhee's review surveys this modeling lineage (McPhee 2022).

A caution cuts the other way: optimal control optimizes *within a model*, and the sensitivity of optimal timing to model structure (hub mobility, shaft flexibility (Betzler et al. 2012; Osis and Stefanyshyn 2012), actuator rate limits) is exactly why a multi-engine platform with cross-engine validation (Section 5) is the right instrument for revisiting the question.

2.5 What "energy transfer" means operationally

Claims about energy transfer require an accounting framework. Four levels of analysis appear in the literature, and they are not interchangeable.

2.5.1 Joint power and work

For a joint spanning segments i and $i+1$ with net joint torque τ and relative angular velocity ω, the *joint (moment) power* is $P_m = \tau\,\omega$; its time integral over a phase is the mechanical work done by the net moment. Positive work indicates generation, negative work absorption. This is the quantity most golf studies report, and it is the basis of the work-power budget of Nesbit and Serrano (Section 3.1) (Nesbit and Serrano 2005).

2.5.2 Joint-force power and segmental energy accounting

Joint moments are not the only conduit of mechanical energy: the joint reaction force $\mathbf{F}$ acting at a joint moving with velocity $\mathbf{v}$ transmits *joint-force power* $P_f = \mathbf{F} \cdot \mathbf{v}$ from one segment to the next. Robertson and Winter's classic formulation balances, for each segment, the rate of change of segment mechanical energy against the sum of joint-force powers and moment powers at its proximal and distal ends, distinguishing energy *transferred between* segments from energy *generated or absorbed* at joints (Robertson and Winter 1980; Winter 2009). This is the correct level of analysis for the timing question, because "retain-early / release-late" is precisely a claim about the *time course of inter-segment transfer terms*, not about peak angular velocities. Section 6 implements exactly this accounting at the model's wrist interface, including the energy-balance residual check.

2.5.3 Induced accelerations and dynamic coupling

In a coupled chain, any single torque induces accelerations at *every* joint — including joints it does not span — through the inverse inertia matrix (Zajac and Gordon 1989). Induced-acceleration and power-transfer analyses in locomotion research (Zajac, Neptune, and Kautz 2002) show that intuitions of the form "muscle X accelerates joint Y" routinely fail in coupled systems. For golf this matters because a torso torque early in the downswing partly accelerates the *club* (through coupling), and a wrist torque late in the downswing back-reacts on the torso; the net energy routing is an emergent property of the whole chain. The ZTCF/ZVCF decompositions of Section 4 are the counterfactual cousins of this analysis.

2.5.4 Kinematic sequence as a proxy — and its failure modes

Peak-ordering of segment angular speeds is a *proxy* for energy flow, not a measurement of it. Marsan and colleagues showed that the identified kinematic sequence in golf depends materially on which angular-velocity component (or norm, or projection) is analyzed: thirteen golfers analyzed with seven component-selection methods yielded different sequences (Marsan et al. 2019). Marshall and Elliott's long-axis-rotation caution (Marshall and Elliott 2000) is a special case of the same problem, and the near-planarity of the downswing is itself only an approximation (Coleman and Rankin 2005; Kwon et al. 2012). Any serious test of the timing hypothesis must therefore run at the level of energy and power accounting (Section 2.5.2), with the kinematic sequence reported as corroborating description only — which is how Section 7 uses it.

Chapter 3

Empirical Evidence in Golfers

3.1 The work and power budget of the swing

Nesbit's full-body kinematic-kinetic study and the companion work-power analysis of Nesbit and Serrano remain the most complete published energy budgets of the swing (Nesbit 2005; Nesbit and Serrano 2005). Using subject-specific full-body models of four golfers of widely differing skill, they report that the trunk complex (lumbar and thoracic spine plus hips) contributes 68.7–72.2% of total swing work, the shoulder and arm joints 24.3–28%, and the legs only 3.3–3.8% (Nesbit and Serrano 2005). Two further observations bear directly on the timing question. First, power generation and clubhead speed decreased systematically with increasing handicap. Second, the phase analysis locates the *generation* of energy predominantly in the earlier downswing (torso-dominated) and the *conversion* into clubhead speed late (shoulder/arm and club interface) — generation and transfer are temporally separated, which is a precondition for the retain-early/release-late strategy to even be possible.

The ground is the ultimate source of the external impulse. Han and colleagues characterized golfer-ground interaction in 63 skilled golfers and found meaningful associations between ground-interaction force and moment parameters and maximum clubhead speed (Han et al. 2019). Recent work has pushed this to an explicitly energetic formulation: in thirty golfers (fifteen professionals, fifteen high-level amateurs), foot-ground interaction variables alone explained 35.5% of clubhead-speed variance, rising to 75.4% when trunk sequencing and an *impulse-based energy-transfer efficiency* measure (effectiveness of mechanical impulse transmission from proximal segments to the club) were added; mediation analysis attributed the dominant indirect pathway to transfer efficiency rather than to sequencing per se ("Foot–Ground Interaction and Clubhead Speed: Impulse-Based Energy Transfer as the Key Mechanism in the Golf Swing" 2026). That distinction — transfer *efficiency* mattering where sequence *order* alone does not — recurs throughout the skill-level literature below.

3.2 Segmental kinetic-energy sequencing

Two studies measured the quantity this document cares most about — segment *kinetic energy*, not merely angular speed — and both found the P→D structure at the energetic level. Anderson, Wright and Stefanyshyn computed segmental kinetic energies in the golf swing and documented the sequenced build-up and outward migration of energy along the chain (Anderson, Wright, and Stefanyshyn 2006). Kenny, McCloy, Wallace and Otto reproduced segmental kinetic-energy sequencing in a large-scale computer-simulated golf swing, confirming the ordered energetic handoff in a full forward-dynamic setting (Kenny et al. 2008). These studies matter here for two reasons. First, they validate the segment-energy time course as a practical observable (the same observable computed in Section 7 for the UpstreamDrift model). Second, they show the energetic sequence is not an artifact of angular-velocity bookkeeping: energy itself arrives late in the club, after the proximal segments have peaked.

3.3 Kinematic-sequence evidence and skill-level differences

The descriptive P→D findings are robust in skilled populations. Tinmark and colleagues, studying elite golfers across full and partial shots, found a significant proximal-to-distal temporal ordering of peak speeds (pelvis → torso → hands/club) with successive amplification of peak speed at every level of effort, in both men and women (Tinmark et al. 2010). Cheetham and colleagues compared amateur recreational players with touring professionals on transition- and downswing-phase kinematic sequence parameters and found professionals showed higher peak-speed magnitudes and more consistent sequencing and timing characteristics (Cheetham et al. 2008). Zheng and colleagues' three-dimensional comparison of professional and amateur swings found systematic temporal and spatial differences, with amateurs deviating progressively more from professional norms as handicap increased (Zheng et al. 2008). Meister and colleagues provide elite rotational benchmarks (pelvis-thorax separation and its timing) against which amateurs can be compared (Meister et al. 2011), and Lindsay, Mantrop and Vandervoort review the broader catalogue of skill-level biomechanical differences (Lindsay, Mantrop, and Vandervoort 2008).

Torso-pelvis interaction variables predict performance. Myers and colleagues found upper-torso and pelvis rotation (and their separation) associated with ball speed in golfers across skill levels (Myers et al. 2008); Chu, Sell and Lephart identified a small set of biomechanical variables — among them trunk and pelvis kinematics at specific downswing events — explaining a large share of driving-performance variance (Chu, Sell, and Lephart 2010); Joyce and colleagues related three-dimensional trunk kinematics to clubhead speed across clubs (Joyce et al. 2013). On the widely coached "X-factor" specifically, Kwon and colleagues showed the computed values depend materially on methodology and that, in skilled golfers, X-factor parameters were not straightforwardly associated with clubhead velocity (Kwon et al. 2013) — another instance of a popular proxy dissolving under measurement scrutiny.

Three refinements prevent an overly tidy reading:

1. **Sequence order is common but not universal — even among the best.** Skilled golfers whose peak ordering deviates from the canonical pelvis-thorax-arm-club pattern are documented, and recent work examines how alternative sequence patterns among elite players relate to clubhead speed (Lee, Ehrlich, and Cain 2026). On-plane analyses of the "axle-chain" system in 66 skilled golfers found clubhead-speed associations concentrated in specific angular-motion characteristics rather than in gross ordering (Madrid et al. 2020).
2. **Within-player timing differences do not straightforwardly explain shot quality.** Comparing well-timed and mistimed shots *within* highly skilled players, Neal and colleagues found no significant differences in the timing lags between peak angular speeds of contiguous segments (Neal et al. 2007) — the felt sense of "timing" lives somewhere other than the inter-peak lags of a five-segment model. Movement variability is itself structured and functional in golf (Langdown, Bridge, and Li 2012; Glazier 2011), so single-trial sequence snapshots carry limited information.
3. **Method sensitivity.** As Section 2.5 noted, the identified sequence depends on component-selection methodology (Marsan et al. 2019), so cross-study comparisons of ordering statistics carry real uncertainty. Historical measurements with specialized instrumentation reached compatible conclusions about trunk-pelvis motion but with their own method dependence (McTeigue et al. 1994).

What survives these refinements: *skilled golfers generate larger proximal speeds, separate transition from release more distinctly, and transmit impulse and energy to the club more efficiently; merely exhibiting a P→D peak order is neither unique to skill nor sufficient for it* (Cheetham et al. 2008; Zheng et al. 2008; "Foot–Ground Interaction and Clubhead Speed: Impulse-Based Energy Transfer as the Key Mechanism in the Golf Swing" 2026; Lee, Ehrlich, and Cain 2026). Coaching perception has partly internalized this: when surveyed systematically, coaches' key swing parameters overlap with, but do not coincide with, the parameters the biomechanical literature supports (Smith et al. 2015).

3.4 Where the transfer occurs

Synthesizing Section 3.1, Section 3.2 and Section 3.3 with the mechanics of Section 2: in a skilled swing, the dominant energy *generation* occurs in the trunk-hip complex during the first two-thirds of the downswing (Nesbit and Serrano 2005); the dominant *transfer to the club* occurs in the final phase, through the wrist interface, by a mechanism that is substantially parametric/geometric (wrist-cock release under centrifugal loading, hub-path tightening) with a modest late active wrist contribution (White 2006; Miura 2001; Springings and Neal 2000; Nesbit and McGinnis 2009). Less-skilled players truncate exactly this separation: early release ("casting") moves energy into the club before proximal speed has peaked, producing the lower distal peak speeds and transfer ratios observed in amateur cohorts (Cheetham et al. 2008; Zheng et al. 2008; Milburn 1982). Chapter Section 7 reproduces this entire pattern — including the casting failure mode — inside a fully instrumented model where every energy pathway is measurable exactly.

Chapter 4

The Timing Question and the Counterfactual Framework

4.1 Two strategies, stated precisely

Let $E_p(t)$ be the kinetic energy of the proximal subsystem (trunk and arms, with the club riding folded) over a downswing $t \in [t_0, t_{\text{imp}}]$, and define the transfer functional $P_{\text{w}}(t)$ as the inter-segment power crossing the wrist interface — joint-force power plus moment power, in the sense of Section 2.5.2. The two strategies are:

- **S1 (transfer early):** make $P_{\text{w}}(t)$ large as soon after transition as possible; the club begins acquiring its final energy early.
- **S2 (retain early, release late):** keep $P_{\text{w}}(t)$ small early (the wrist transmits centripetal force but little power) while proximal torques maximize E_p; then allow or drive a large P_{w} concentrated in the final phase.

The double-pendulum, delayed-release and optimal-control literatures reviewed in Section 2 uniformly favor S2 within their model classes (Milburn 1982; Pickering and Vickers 1999; Sprigings and Neal 2000; Sprigings and MacKenzie 2002; Sharp 2009; White 2006). The mechanistic reasons compose into a coherent account:

1. **Inertia management.** With the wrist cocked, the system's moment of inertia about the hub is small, so proximal torque buys angular velocity cheaply; early release raises the inertia while proximal torque is still accelerating the system (White 2006; T. P. Jorgensen 1999).
2. **Actuator physiology.** Torque-velocity limits mean the wrist's small muscles add meaningful power only while wrist angular velocity is modest; early wrist torque squanders that window and can even require *negative* (restraining) torque later (Sprigings and Neal 2000).
3. **Parametric transfer needs developed centrifugal loading.** The passive outward energy flow at release, and the inward-pull gain, scale with the centrifugal field established by proximal rotation — weak early, strong late (White 2006; Miura 2001).
4. **Interaction torques do the late work for free.** Motion-dependent torques accelerate the distal segment and decelerate the proximal one automatically once release begins; the golfer's job is to *time* this cascade, not to muscle it (Putnam 1993; Herring and Chapman 1992).

Empirically, S2-like signatures separate skill groups (Section 3.3), while the within-expert null result on inter-peak lags (Neal et al. 2007) suggests that once S2 is grossly achieved, residual performance variance lives in finer variables (release profile shape, hub-path geometry, impulse-transfer efficiency) — exactly what the counterfactual framework below is designed to expose.

4.2 What would falsify S2?

To keep the question scientific rather than rhetorical, we state what evidence would count against the retain-early/release-late hypothesis:

- an optimal-control solution, in a validated 3D model with physiological actuator limits, whose optimal wrist-interface power is front-loaded;
- measured expert swings in which early wrist-interface power (normalized) is *positively* associated with clubhead speed across players;
- counterfactual (ZTCF/ZVCF) analyses showing that in faster swings the control term dominates club acceleration *early* rather than late;
- **or, in the direct manipulation of Section 6: torque programs that drive the wrist early outperforming programs that retain early and drive late, under matched proximal effort.**

The last test is executed in Section 7.

4.3 ZTCF and ZVCF: definitions

Two counterfactual accelerations are defined *pointwise* at each measured state $(q, \dot{q}, \tau)$ of a swing:

$$\ddot{q}_{\mathrm{ZTCF}} \;=\; M(q)^{-1}\,(-\,b(q,\dot{q})), \tag{4.1}$$

$$\ddot{q}_{\mathrm{ZVCF}} \;=\; M(q)^{-1}\,(\tau \;-\; b(q,\mathbf{0})) \;=\; M(q)^{-1}\,(\tau \;-\; g(q)), \tag{4.2}$$

where the simplification in Equation 4.2 holds because the Coriolis term (quadratic in velocity) and viscous dissipation (linear in velocity) vanish at $\dot{q} = \mathbf{0}$. The ZTCF (Equation 4.1) answers *"what would the system do right now if every actuator switched off?"* — it is the **drift** field of the affine control system, containing gravity plus all motion-dependent interaction dynamics. The ZVCF (Equation 4.2) answers *"what would the currently applied torques, together with gravity, produce from a standstill in this pose?"* — it removes every motion-dependent term but is *not* the pure control contribution, since gravity remains in Equation 4.2; the pure torque term is $M^{-1}\tau$ in the split below. Because Equation 2.1 is affine in τ, the actual acceleration superposes exactly:

$$\ddot{q} \;=\; \underbrace{M^{-1}(-b(q,\dot{q}))}_{\text{drift = ZTCF}} \;+\; \underbrace{M^{-1}\tau}_{\text{control}}, \tag{4.3}$$

an identity enforced to numerical tolerance by the platform's test suite (**?@sec-appendix-validation**).

4.4 Two operational variants

The platform implements two distinct operationalizations, and the difference matters for interpretation.

Pointwise (instantaneous) counterfactuals. The engine-agnostic Python implementation (`simulation_backends/ztcf_` evaluates Equation 4.1 and Equation 4.2 at each sample of an already-measured trajectory — explicitly *not* a forward-integrated rollout. Each sample answers a local "what if"; this is the right tool for drift/control *time profiles* along a real swing, and it is the tool used in Section 7.

Re-simulation counterfactuals. The original MATLAB/Simscape pipeline (`run_ztcf_simulation.m`) re-simulates the full model with a torque "killswitch" applied at time t, for a sweep of t values, tabulating the counterfactual state reached with torques off. The pipeline forms the difference table DELTA = BASE−ZTCF (`process_data_tables.m`), attributing to active control everything the passive system would not have done. A companion ZVCF model applies the measured torques to the same pose with velocities zeroed (`SCRIPT_ZVCF_GENERATOR.m`); the ZVCF model is run without gravity so that, when its samples are set against the ZTCF (which retains gravity), the gravitational contribution is not double-counted. This decomposition is meaningful *pointwise*, at matched states, in the sense of Equation 4.3: the ZVCF script records one sample per zero-velocity pose, and separately integrated counterfactual rollouts do not superpose at the trajectory level in a nonlinear multibody system, because M and b evolve differently once the counterfactual state diverges. The MATLAB pipeline additionally computes, for every joint and every table (BASE, ZTCF, DELTA), the linear work $\int \mathbf{F} \cdot \mathbf{v} \, dt$, angular work $\int \tau \cdot \omega \, dt$, total work and power, and the fractional contributions ZTCF/BASE and DELTA/BASE (`calculate_work_impulse.m`, `calculate_total_work_power.m`) — precisely the Robertson-Winter-style joint-interface accounting of Section 2.5.2, including a kinematic-sequence plot computed *separately for the BASE, ZTCF and DELTA components* (`SCRIPT_324_PLOT_ZTCF_KinematicSequence.m`).

4.5 The counterfactual form of the timing hypothesis

The ZTCF/ZVCF language makes the S2 hypothesis crisp. Along a measured downswing, project the drift and control accelerations onto the club's speed-producing direction and integrate the associated powers. S2 predicts, for high-performing swings:

- **Early downswing:** the *control* term should dominate energy input at proximal joints, while at the wrist the control contribution to club angular acceleration should be small or negative (restraining lag against the developing centrifugal field). The club's ZTCF profile should show the drift field already "wants" to open the wrist — control is *holding energy back*.
- **Late downswing:** the *drift* term (parametric transfer, centrifugal release) should dominate club acceleration, with a brief late window of positive wrist control power (Sprigings and Neal 2000). The active share of club-side work should concentrate before release; the passive share should account for most club energy gain after release (White 2006).

Before this thesis, this pattern was a hypothesis motivated by the platform's preliminary MATLAB analyses. Section 7 now provides the first committed, reproducible quantitative test on the platform's two-segment model; conversely, had the early-drive programs won the sweep, S2 would have been falsified in exactly the terms of 3. The remaining generality — re-simulation parity, 3D full-body models, and measured swings — is future work tracked in issues #8377 and #8378.

Chapter 5

The UpstreamDrift Platform

This chapter summarizes the platform capabilities the modeling pathway builds on; the complete hot-linked inventory, including known gaps, is in 4.

5.1 Models

Table 5.1: Golf swing models available in the repository.

Model	Location	Notes
Double pendulum (arms + club, clubhead tip mass)	`pendulum_simulator/physics.py`	Lagrangian dynamics; viscous + Coulomb friction; analytic M, C, g
Backend-neutral golf model (ODE / MuJoCo / MJX / Warp)	`simulation_backends/`	One parameter set rendered to every backend; shared trace schema; **used in Section 6**
Triple pendulum	`pendulum_simulator/physics_triple.py`	Three-segment planar chain
8-DOF closed-loop golfer (two arms + shared club)	`pendulum_simulator/physics_golfer.py`	Holonomic loop constraints; JAX variant
MuJoCo humanoid golfer	`mujoco_humanoid_golf/`	Full-body model with club
Simscape 2D/3D golf models	`Simscape_Multibody_Models/`	Source of the original ZTCF/ZVCF methodology
Flexible shaft (FEM)	`physics/flexible_shaft.py`	Beam model; strain energy not yet in the energy budget

5.2 The counterfactual and energetics stack

The normative contract lives in the engine-core dynamics interface (`_dynamics_interface.py`): `compute_drift_acceleration`, `compute_control_acceleration`, `compute_ztcf`, `compute_zvcf`, with the superposition identity (Equation 4.3) stated as a contract. Engine implementations exist for MuJoCo, Drake, Pinocchio, OpenSim, MyoSuite, the analytic pendulum engines, and the Simscape bridge (`engine_capabilities.md`). On top of these sit the pointwise engine-agnostic evaluator with HDF5 persistence (`ztcf_zvcf.py`), a MuJoCo delta-based counterfactual analyzer (`counterfactuals.py`), induced-acceleration analyzers, MuJoCo inter-segment power flow (`power_flow.py`), engine-agnostic work/power/impulse metrics (`power_work_metrics.py`), kinematic-sequence and coordination analytics (`kinematic_sequence.py`, `coordination_metrics.py`), and the MATLAB BASE/ZTCF/DELTA/ZVCF pipeline of Section 4.4.

5.3 Why this platform is the right instrument

Three properties make UpstreamDrift suited to the timing question. First, **one model, many engines**: the same `GolfModelParams` specification renders to an analytic CPU reference backend and to MuJoCo-family backends, so results can be cross-validated between independent dynamics implementations (**?@sec-appendix-validation**). Second, **counterfactuals as first-class citizens**: the drift/control split is a tested contract, not an ad-hoc analysis, so the S2 hypothesis of Section 4.5 can be evaluated with the same code on every engine. Third, **reproducibility discipline**: the experiments in Section 6 are committed modules with recorded provenance, run by two commands (**?@sec-appendix-repro**), and their outputs are the only source of every number in Section 7.

Chapter 6

Methods: The Executed Modeling Pathway

This chapter specifies the simulation experiments whose results appear in Section 7. Everything described here is implemented in three committed modules under `scripts/research/proximal_distal_energy/` (`swing_model.py`, `torque_programs.py`, `run_experiments.py`, with figures from `make_figures.py`), and every output carries a provenance record (git SHA, integrator settings, parameter values). The reproduction commands are in **?@sec-appendix-repro**.

6.1 Model

The plant is the platform's backend-neutral two-segment golf model (Section 5): an arm segment rotating about a fixed hub, hinged at the wrist to a club segment consisting of a distributed shaft plus a clubhead point mass at the tip, moving in an inclined swing plane with projected gravity. Parameters are the repository defaults (`GolfModelParams.default()`; full table in **?@sec-appendix-params**): arm length 0.75 m and mass 7.5 kg; club length 1.0 m with 0.15 kg shaft and 0.20 kg clubhead; plane inclination 35°; viscous damping 0.4 and 0.25 $\mathrm{N \cdot m \cdot s/rad}$ at shoulder and wrist. Angles follow the platform convention: θ_1 is the arm angle from the downward vertical in the swing plane, θ_2 the wrist angle of the club relative to the arm, and the club's absolute angle is $\theta_{\text{club}} = \theta_1 + \theta_2$.

Rollouts use the ODE reference backend (analytic dynamics, fixed-step RK4) with $\Delta t = 1$ ms over a 0.9 s horizon. Every rollout starts from the same top-of-backswing state, $\theta_1(0) = -2.2$ rad, $\theta_2(0) = -1.57$ rad (a 90° wrist cock), $\dot{q}(0) = 0$ — an arm carried well past horizontal with the club folded over the shoulder, comparable to the classical two-lever setups (T. P. Jorgensen 1999; Pickering and Vickers 1999).

6.2 Torque programs

All programs share a constant shoulder torque τ_s active from $t = 0$ (two levels: 60 and 100 $\mathrm{N \cdot m}$), so the sweep isolates the wrist channel. The wrist torque follows one of three profiles, parameterized by an onset time t_{on}:

- **passive**: $\tau_w(t) = 0$ throughout (free hinge) — the White limit (White 2006);
- **drive-only**: $\tau_w(t) = 0$ for $t < t_{\text{on}}$, then $\tau_w = +15$ $\mathrm{N \cdot m}$ (opening);
- **restrain-then-drive**: $\tau_w(t) = -R$ (cock-retaining) for $t < t_{\text{on}}$ with $R \in \{5, 10\}$ $\mathrm{N \cdot m}$, then $\tau_w = +15$ $\mathrm{N \cdot m}$.

The onset grid is $t_{\text{on}} \in \{0, 0.025, \ldots, 0.35\}$ s (15 values). With both shoulder-torque levels this yields 92 distinct programs; `drive-only` at $t_{\text{on}} = 0$ is the S1 ("transfer early") pole of the design and `restrain-then-drive` with late onset is the S2 pole. Torque magnitudes sit inside the ranges used by comparable two- and three-segment studies (Sprigings and Neal 2000; Pickering and Vickers 1999), scaled to this model's inertias.

E1b (actuator-bound sensitivity). Because constant torques overstate what muscle can deliver at speed, a companion sweep (`e1b_bounded_torque.py`) replaces both concentric drives with a linear torque-velocity bound, $\tau(\omega) = \tau_{\text{iso}}\,\text{clip}(1 - \omega/\omega_{\text{max}}, 0, 1)$ (shoulder: $\tau_{\text{iso}} = 100$ N $\cdot$ m, $\omega_{\text{max}} = 20$ rad/s against arm speed; wrist: 20 N $\cdot$ m, 30 rad/s against opening speed; restraint constant, since eccentric torque does not taper with shortening velocity). State-dependent torques require closed-loop integration, done with the backend's own `mass_matrix`/`bias_forces` primitives under the identical RK4 settings.

6.3 Impact definition and validity

Impact is the first upward crossing of $\theta_{\text{club}} = 0$ (club pointing straight down in the swing plane), linearly interpolated between samples; the score of a program is the clubhead (distal tip) speed at that instant. Because the shoulder torque is constant, a program that keeps the club folded through the bottom of the swing can postpone the crossing into a second rotation while the shoulder keeps pumping energy — an artifact, not a swing. A trial is therefore **invalid** unless the arm angle at the crossing satisfies $\theta_1 \leq 2.0$ rad (a first-pass delivery); 29 of the 92 programs were excluded by this rule, all of them heavy-restraint or very-late-onset programs. This constraint is conservative *against* the S2 hypothesis: it discards only trials that would otherwise have inflated the restraint arm of the sweep.

6.4 Recorded quantities

For every program: the full state trajectory, applied controls, impact time and clubhead speed. For four representative programs at $\tau_s = 60$ N $\cdot$ m (passive; drive-only at $t_{\text{on}} = 0$; the best drive-only; the best restrain-then-drive), additionally:

Segment energies. Arm kinetic energy $\frac{1}{2}I_1^{\text{hub}}\dot\theta_1^2$; club kinetic energy $\frac{1}{2}m_2\|\mathbf{v}_{c2}\|^2 + \frac{1}{2}I_2^{\text{com}}(\dot\theta_1 + \dot\theta_2)^2$ with the club centre-of-mass velocity $\mathbf{v}_{c2}$ computed analytically from the state.

Wrist-interface power accounting (E4). Following Section 2.5.2: the joint-force power $P_f = \mathbf{F} \cdot \mathbf{v}_{\text{wrist}}$, with the joint reaction force recovered from Newton's second law for the club segment, $\mathbf{F} = m_2\mathbf{a}_{c2} - m_2\mathbf{g}$ (centre-of-mass acceleration by central differences of the analytic velocity); the net wrist moment power on the club $(\tau_w - b_w\dot\theta_2)(\dot\theta_1 + \dot\theta_2)$; the wrist actuator power $\tau_w\dot\theta_2$; and the residual check that the club's total mechanical energy rate equals joint-force power plus moment power (reported as a figure in Section 7; gravity is not added separately because potential energy is included in the club's mechanical energy).

Pointwise counterfactual split (E2). At every sample, the platform's `drift_and_control_split` (Equation 4.3) evaluated with the applied torques, yielding drift and control components of both joint accelerations; the club's absolute angular-acceleration split is the row sum.

Phase budgets. With the downswing divided at $t_{\text{imp}}/2$: club kinetic-energy gain, wrist actuator work, shoulder work, and integrated joint-force transfer, per half.

6.5 Design rationale and scope

This is deliberately the *simplest* member of the platform's model family (Table 5.1): a planar 2-DOF chain with a fixed hub and rigid shaft. That choice is the point, not a compromise: the classical timing results were established in exactly this model class (Section 2.3), so the pathway first reproduces the class's behavior under the platform's instrumentation — where every energy pathway is exactly measurable and the counterfactual contract is tested — before scaling to models where the same questions are harder to instrument. What the 2-DOF results can and cannot establish is taken up explicitly in Section 8; the extension pathway (re-simulation parity, 8-DOF golfer, full-body engines, motion capture) is specified in Section 9 and tracked in issues #8377 and #8378.

Chapter 7

Results

All numbers in this chapter are computed by the committed experiment suite (Section 6) from the recorded outputs in `data/` (`e1_sweep.json`, `representative_traces.npz`, `results_summary.json`); figures are generated by `make_figures.py`. Speeds are quoted to 0.1 m/s and times to 1 ms.

7.1 E1 — Timing of the distal handoff

Figure 7.1 shows clubhead speed at impact as a function of wrist-drive onset time for both shoulder-torque levels. Table 7.1 lists the representative programs.

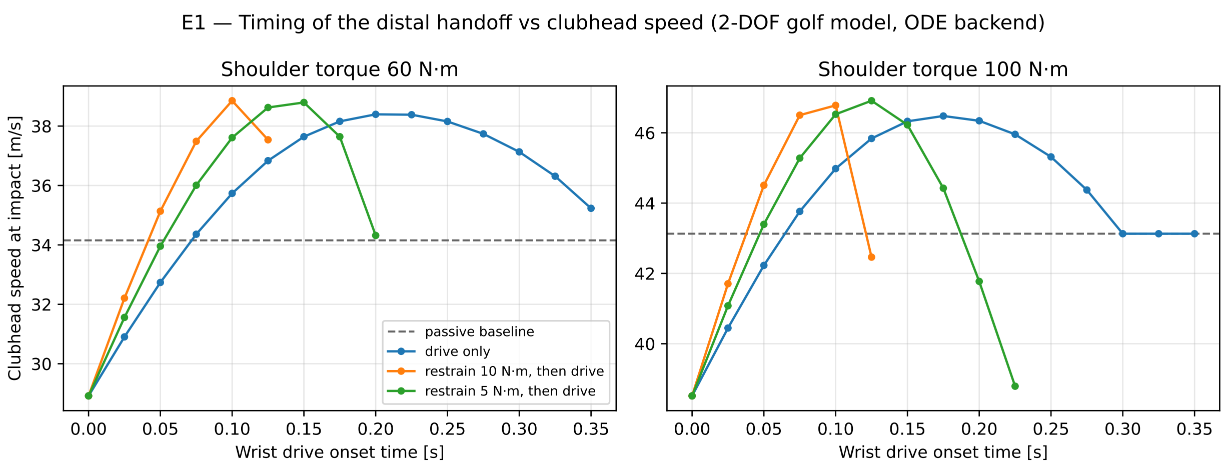

Figure 7.1: E1 sweep: clubhead speed at impact versus wrist-drive onset time, for drive-only and restrain-then-drive profiles at two shoulder-torque levels. Dashed line: fully passive wrist baseline.

Table 7.1: Representative programs at $\tau_s = 60$ N $\cdot$ m.

Program ($\tau_s = 60$ N $\cdot$ m)	t_{on} [s]	t_{imp} [s]	Clubhead speed [m/s]	vs passive
Passive wrist	—	0.370	34.2	—
Early drive (S1 pole)	0.000	0.327	28.9	**−15.3%**
Best drive-only	0.200	0.352	38.4	**+12.4%**

Program ($\tau_s = 60$ N · m)	t_{on} [s]	t_{imp} [s]	Clubhead speed [m/s]	vs passive
Best restrain-then-drive ($R = 10$)	0.100	0.349	38.9	**+13.8%**

Three findings, consistent across both torque levels:

1. **Driving the wrist from the top is the worst valid strategy.** At $\tau_s = 60$ N · m the early-drive program reaches 28.9 m/s versus 34.2 m/s for a completely passive wrist — active early distal effort *destroys* 15% of the speed a free hinge would have delivered. The early-drive program is the worst valid trial in the entire 92-program sweep. At $\tau_s = 100$ N · m the pattern repeats (early drive is the minimum of its profile curve; passive baseline 43.1 m/s).
2. **The same wrist torque, applied late, is worth +12%.** Clubhead speed rises monotonically with onset time until an interior optimum ($t_{\text{on}} = 0.200$ s at 60 N · m, 0.175 s at 100 N · m — roughly the middle of the downswing, with drive active through the final ~40%), reaching 38.4 and 46.5 m/s respectively.
3. **Retain-early-then-drive is best of all.** Programs that apply a *negative* (cock-retaining) wrist torque early and the identical positive drive late beat the best drive-only programs at both torque levels: 38.9 m/s ($R = 10$ N · m until 0.100 s) at $\tau_s = 60$, and 46.9 m/s ($R = 5$ N · m until 0.125 s) at $\tau_s = 100$. The margin over the best drive-only program is modest (+0.5 m/s at both levels), consistent with most of the retention benefit being achievable passively; but the ordering is strict, and heavy or over-long restraint fails the first-pass validity rule

    (2) rather than winning — the benefit is a *timed* retention, not maximal retention.

7.2 E1b — Sensitivity to actuator torque-velocity bounds

A standing objection to constant-torque sweeps is that real muscle torque falls with shortening velocity, shrinking the late-drive window (Sprigings and Neal 2000). E1b re-runs the sweep with a linear torque-velocity bound on both concentric drives (shoulder: 100 N · m isometric, zero at 20 rad/s of arm speed; wrist: 20 N · m isometric, zero at 30 rad/s of opening speed; restraint left constant, as eccentric torque does not taper with shortening velocity), integrated closed-loop with the same RK4 settings (Figure 7.2). The ordering survives intact: early drive 28.8 m/s < passive 30.8 < best late drive 34.6 (onset 0.175 s) < best restrain-then-drive 34.8 (R = 10 N · m until 0.100 s). Effect sizes compress — late drive is worth +12.2% over passive and the restraint margin narrows to +0.2 m/s — precisely the compression bounded-actuator theory predicts, while the optimal onsets move slightly *earlier* (0.175–0.100 s versus 0.200–0.100 s), since a velocity-limited wrist needs a slightly longer window to deliver its work. The S2 ordering is therefore not an artifact of unbounded actuators.

7.3 Kinematic sequence

Figure 7.3 shows arm and club angular-speed traces. All four programs produce the qualitative P→D pattern — arm speed peaks, then club speed peaks higher — as expected from interaction dynamics alone (Section 2.2). The programs differ exactly where the skill literature says they should (Section 3.3): the early-drive swing opens the club immediately, its arm peak is depressed, and its club peak is the lowest of the four; the retain-then-drive swing holds the club folded longest, achieves the highest arm speed before release, and converts it into the highest club peak. Impact in the early-drive swing occurs with the arm still 0.36 rad *short* of vertical — the clubhead has overtaken the hands before the arm arrives, the model's version of "casting" / premature release — whereas the two best programs deliver impact with the arm 0.7–0.8 rad *past* vertical (hands ahead of the clubhead, forward shaft lean), matching the geometry of skilled impact.

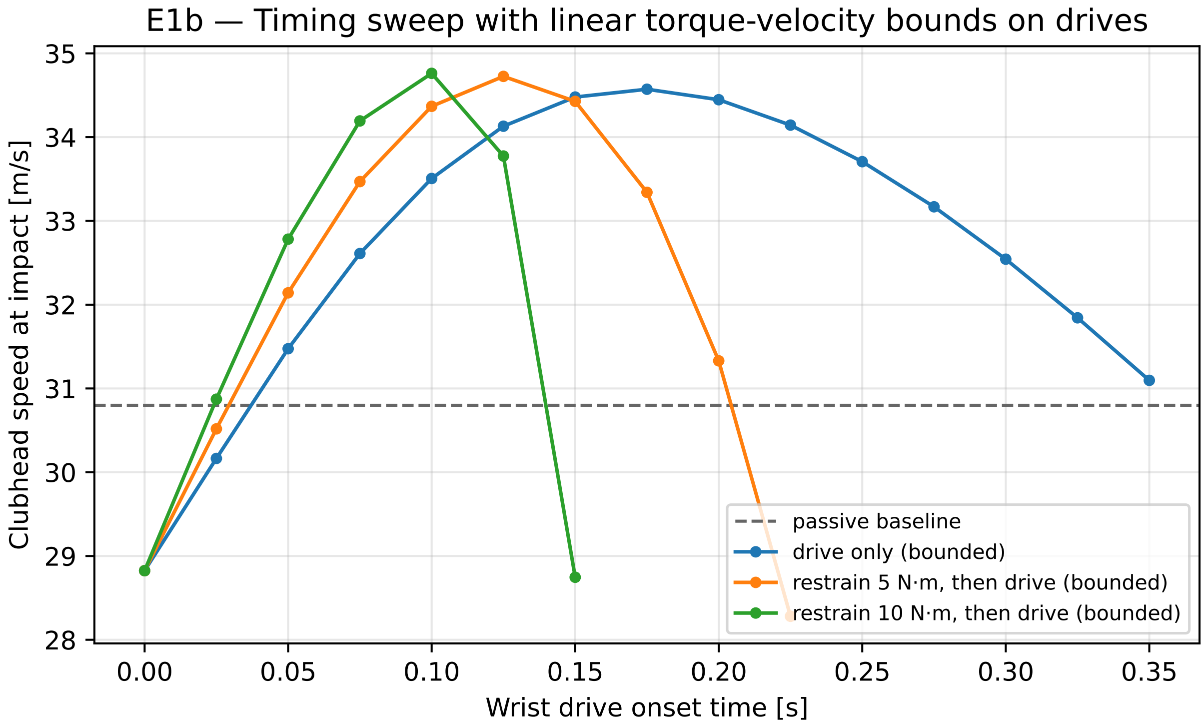

Figure 7.2: E1b: the timing sweep repeated with linear torque-velocity bounds on the shoulder and wrist drives.

7.4 Segment energies

Figure 7.5 shows arm and club kinetic-energy time courses, and Figure 7.6 the early-half/late-half budget. The energetic story is unambiguous:

- **Early half.** The winning programs put the *least* energy into the club early: 5.0 J (best restrain) and 7.3 J (best drive-only) versus 22.9 J for the early-drive program (passive: 7.8 J). Meanwhile they bank more energy in the arm: early-half shoulder work is 54.7 J (best restrain) and 51.9 J (best drive) versus 33.9 J (early drive) — the early-driven club back-reacts on the arm and *steals its runway*.
- **Late half.** The ordering inverts: late club kinetic-energy gain is 180.5 J (best restrain) and 174.2 J (best drive) versus 84.4 J (early drive) and 136.4 J (passive).
- **Active retention is measurable.** The best restrain program's early-half wrist actuator work is *negative* (−1.9 J) and its early-half joint-force transfer is likewise slightly negative (−1.0 J): the wrist interface is momentarily a net energy *withdrawer* from the club while the proximal system loads — the literal, quantitative meaning of "retaining energy proximally".

7.5 E4 — Wrist-interface power accounting

Figure 7.7 (left) shows the Robertson-Winter interface powers for the best restrain-then-drive swing. The joint-force power — energy carried into the club *through the joint force*, the pathway that requires no wrist torque at all — is small and briefly negative early, then surges in the final ~0.1 s to dominate the club's energy supply (late-half integral 131.7 J versus 25.7 J of late wrist actuator work). The wrist moment channel contributes a brief late pulse, exactly the "latter stages" window identified by Sprigings and Neal (Sprigings and Neal 2000). The right panel verifies the accounting: the club's mechanical-energy rate tracks the sum of joint-force and moment power across the swing (residual at the level of the finite-difference approximation),

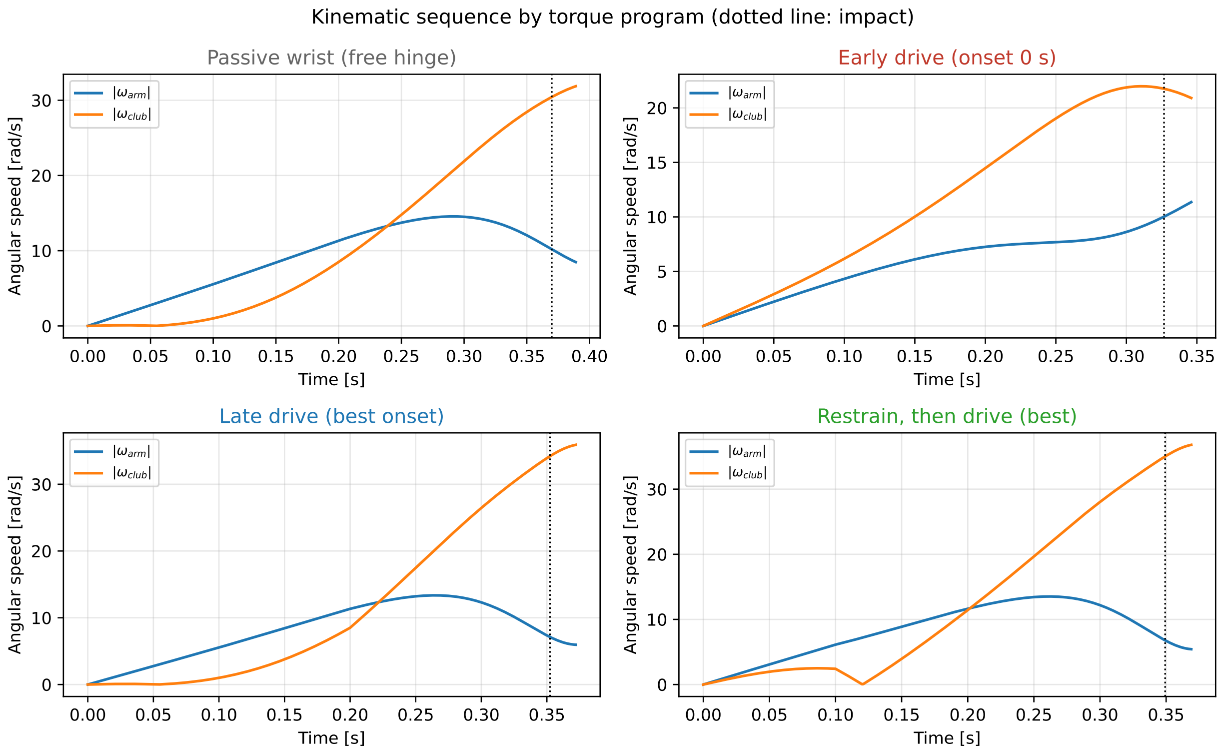

Figure 7.3: Kinematic sequence (arm and club angular speeds) for the four representative programs at $\tau_s = 60$ N · m. Dotted vertical line: impact.

confirming the implementation of the segmental balance of Section 2.5.2.

7.6 E2 — Counterfactual drift/control split

Figure 7.8 shows the pointwise drift/control decomposition (Equation 4.3) of the club's absolute angular acceleration for the S1 pole (early drive) and the S2 winner (restrain-then-drive). In the early-drive swing, the control term supplies a large share of club acceleration from the first instant — the S1 signature — and the drift term never develops the late surge seen elsewhere. In the restrain-then-drive swing the picture reverses: early, the control term *opposes* the drift term at the club (restraint holding back a drift field that already wants to release); late, the drift term — the ZTCF, i.e. what the accumulated motion would do with every actuator switched off — dominates the club's acceleration entirely, with the control term a small additive assist. This is, instant by instant, the counterfactual signature that Section 4.5 predicted for high-performing swings: **early control retains; late drift delivers.**

7.7 Summary of findings

Within this model class, under matched proximal effort and a consistently defined first-pass impact: (i) early distal drive is strictly and substantially harmful (−15% versus doing nothing at all); (ii) identical distal effort applied late is worth +12%; (iii) timed active retention followed by late drive is best at both effort levels, with its early-phase wrist work measurably negative; (iv) the late club energy supply is dominated by the passive joint-force/parametric pathway, not the wrist actuator; (v) the counterfactual split shows control acting early as restraint and drift delivering late — the S2 pattern of Section 4.5 in full; and (vi) the entire ordering survives linear torque-velocity bounds on the drives, with compressed effect sizes and slightly earlier optima (Section 7.2).

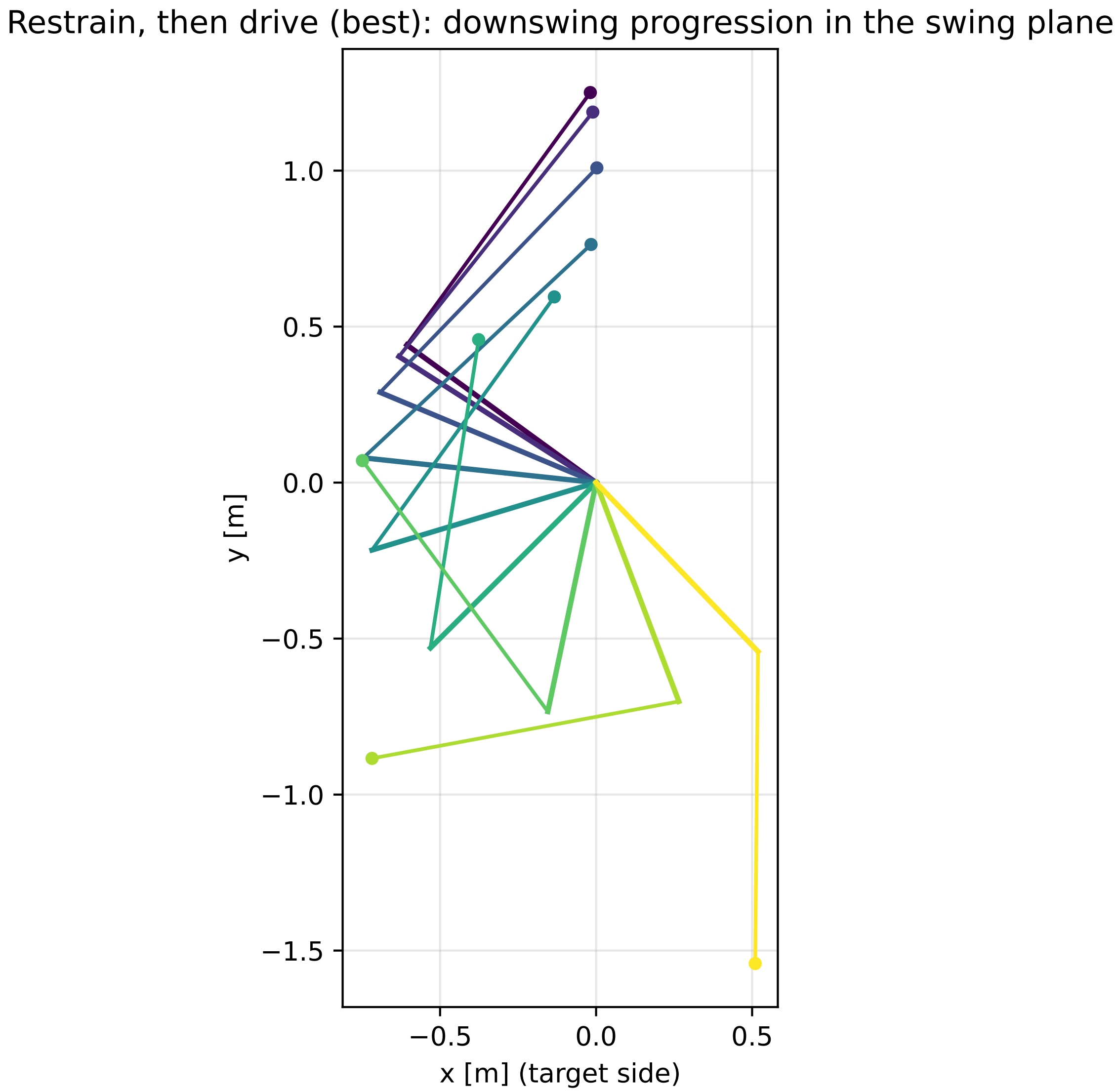

Figure 7.4: Stick-figure montage of the best restrain-then-drive swing in the swing plane (downswing progression at equal time steps).

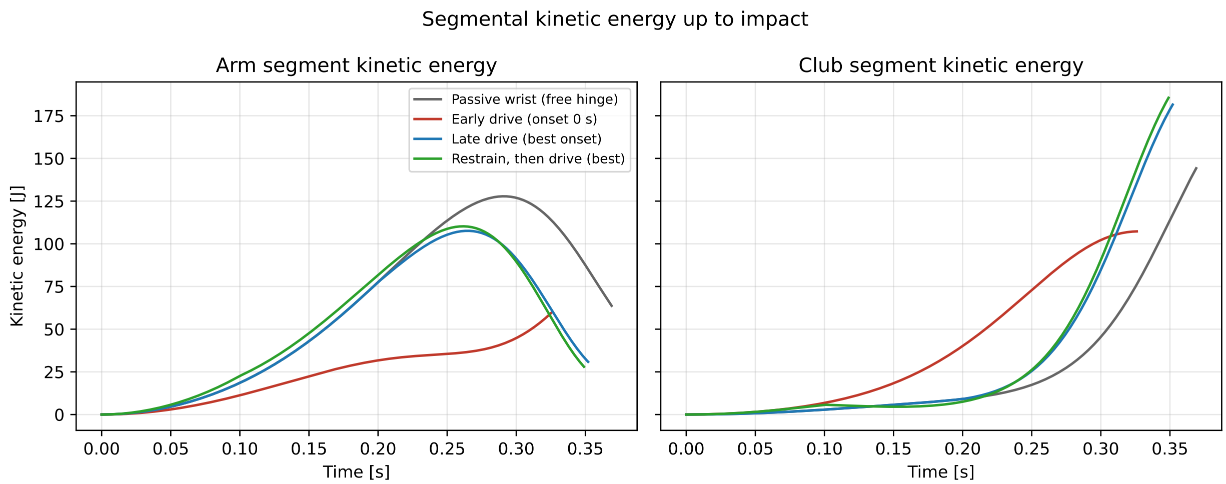

Figure 7.5: Segmental kinetic energies up to impact for the four representative programs.

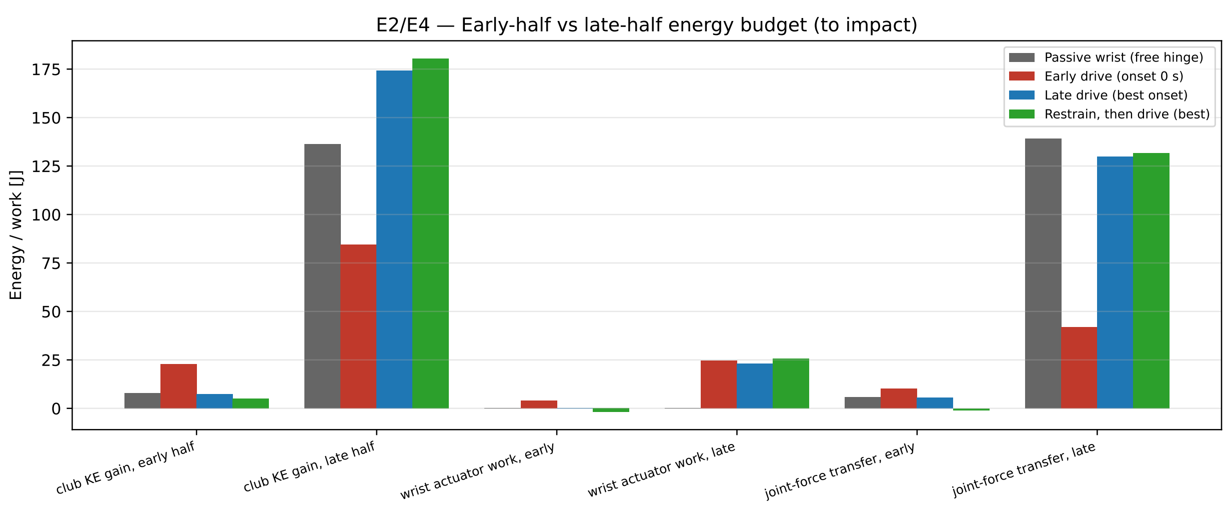

Figure 7.6: Early-half versus late-half energy budget (club kinetic-energy gain, wrist actuator work, integrated joint-force transfer) for the representative programs.

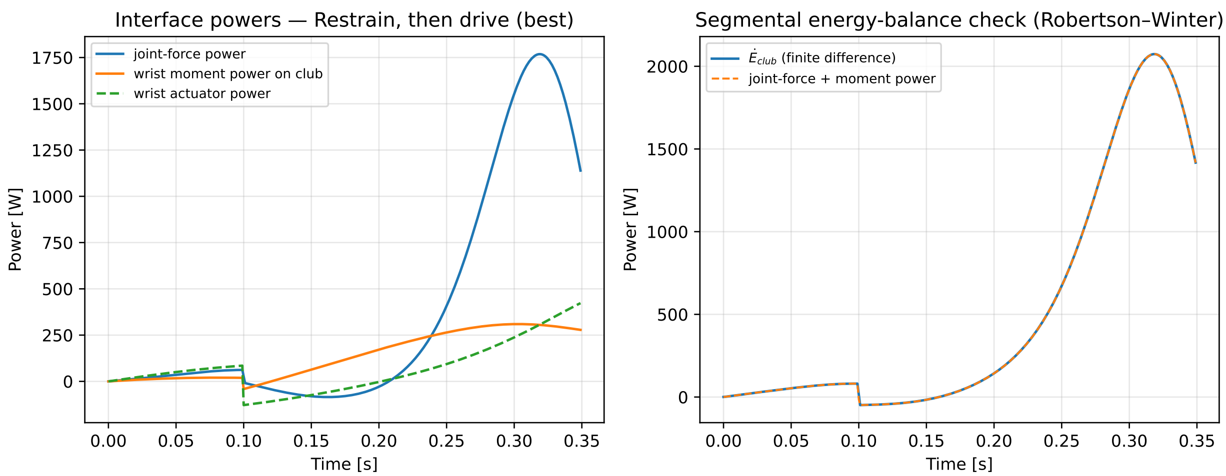

Figure 7.7: Left: wrist-interface powers for the best restrain-then-drive swing. Right: segmental energy-balance check — club mechanical-energy rate versus the sum of interface powers.

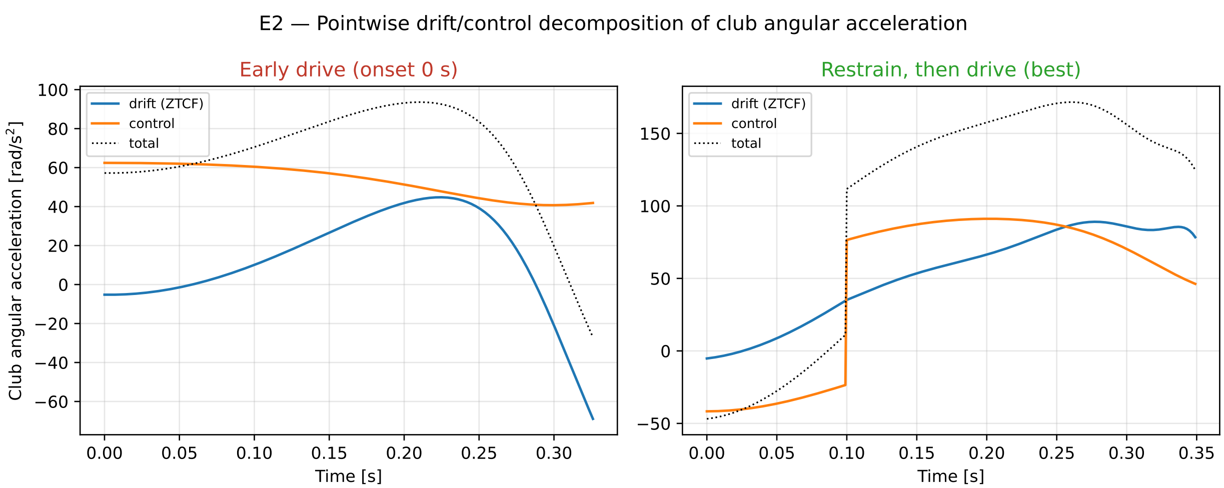

Figure 7.8: Pointwise drift (ZTCF) versus control decomposition of club absolute angular acceleration, for the early-drive (left) and best restrain-then-drive (right) programs.

Chapter 8

Discussion

8.1 Agreement with the literature

Every headline result of Section 7 has an antecedent in the literature of Section 2, and the quantitative agreement is close enough to be meaningful rather than merely directional. The harm of early release reproduces the delayed-release results of Pickering and Vickers and of Sprigings and MacKenzie (Pickering and Vickers 1999; Sprigings and MacKenzie 2002) and Milburn's original observation that early uncocking truncates arm acceleration (Milburn 1982). The +12% value of late wrist drive brackets Sprigings and Neal's 4% gain from a physiologically bounded late wrist torque (Sprigings and Neal 2000) — our unbounded constant 15 $\mathrm{N \cdot m}$ torque and longer active window make a larger gain expected. The dominance of the passive late energy pathway (joint-force transfer of 131.7 J versus 25.7 J of late wrist work in the best swing) is White's parametric transfer made explicit in Robertson-Winter accounting (White 2006; Robertson and Winter 1980). The casting geometry of the early-drive swing — clubhead overtaking the hands before the arm reaches vertical — is the model's rendering of the amateur release failure documented across skill-level studies (Cheetham et al. 2008; Zheng et al. 2008), and the impact geometry of the winning programs (hands ahead, arm 0.7–0.8 rad past vertical) matches the skilled pattern. Finally, the drift/control time signature — control as early restraint, drift as late delivery — is the counterfactual pattern the platform's ZTCF work hypothesized (Section 4.5), now exhibited by the model itself.

8.2 Why retention wins: a mechanistic reading

The energy budgets localize the causal structure. Early wrist drive fails through *two* simultaneous channels: it raises the system's moment of inertia about the hub while the shoulder is still accelerating (so the same shoulder torque buys less arm speed — the early-drive program banks 21 J less early shoulder work than the retain program despite identical τ_s), and it spends the wrist's work budget at a time when the joint-force/parametric channel cannot yet amplify it (White 2006). Timed restraint wins through the mirror mechanisms: it holds the compact configuration slightly longer than a free hinge would (the drift field already wants to open the wrist — the restraint is fighting centrifugal release, not gravity), banks that margin as arm speed, and then releases into a fully developed centrifugal field where the passive channel converts arm energy to club energy at the maximum rate. The negative early wrist work (−1.9 J) is small in absolute terms but diagnostic: retention is an *energetic withdrawal*, not merely an absence of drive — precisely the "restraining torque" role that appears in Jorgensen's fitted expert swing and in Sprigings-Neal-type actuator solutions (T. P. Jorgensen 1999; Sprigings and Neal 2000; Herring and Chapman 1992).

Two honest qualifications temper the S2 reading. First, the margin of active restraint over the best *passive-then-drive* program is small (+0.5 m/s at both effort levels): in this model most of the retention benefit is available from simply *not driving* the wrist, with active holding adding a final increment. Second, retention is only valuable *timed*: heavier or longer restraint programs failed the first-pass validity rule rather than winning — over-retention postpones delivery past the useful window. "Retain early, release late" is therefore not "retain maximally"; it is *retain until the drift field can pay, then stop retaining.*

8.3 What the 2-DOF model cannot establish

The results chapter demonstrates the S2 pattern in the model class where the classical timing literature lives; it does not yet demonstrate it for a golfer. The specific gaps, each with a tracked remediation:

- **Fixed hub, rigid shaft, planar chain.** The hub-path and inward-pull mechanisms (Miura 2001; Nesbit and McGinnis 2009; Nesbit and McGinnis 2014), shaft dynamics (MacKenzie and Sprigings 2009; Betzler et al. 2012; Osis and Stefanyshyn 2012), and long-axis rotations (Marshall and Elliott 2000) are all outside this model. The platform's 8-DOF closed-loop golfer, flexible-shaft FEM model, and full-body engines (Table 5.1) are the staged extensions (issue #8377).
- **Simplified actuator models.** E1b shows the S2 ordering survives a linear torque-velocity bound on the drives with compressed margins (Section 7.2), but activation-rate limits, history dependence, and eccentric-side detail remain unmodeled; full optimal-control treatments with physiological actuators are the appropriate refinement (Sprigings and Neal 2000; Sharp 2009; McNally and McPhee 2018).
- **One impact definition.** Scoring at club-vertical with a first-pass validity bound is principled but not unique; alternatives (fixed hand position, ball-position sweep as in Pickering and Vickers (Pickering and Vickers 1999)) should be checked for ordering robustness.
- **Pointwise counterfactuals only.** The Python pathway evaluates instantaneous ZTCF profiles; the MATLAB re-simulation variant (killswitch rollouts, DELTA tables) has not yet been reproduced in Python, so the two operationalizations have never been formally compared on matched trajectories (issue #8378).
- **No human data yet.** The decisive observational test — wrist interface power time courses across skill levels from motion capture
  (3) — awaits the platform's mocap pipeline experiments.

8.4 Implications, cautiously stated

For modeling practice, the pathway shows that the timing question is tractable with the platform's existing counterfactual stack, and that energy-level accounting (not peak ordering) is the observable that separates strategies — supporting the methodological argument of Section 2.5 and the transfer-efficiency findings in recent human data ("Foot–Ground Interaction and Clubhead Speed: Impulse-Based Energy Transfer as the Key Mechanism in the Golf Swing" 2026). For coaching interpretation, the results align with instruction that discourages "hitting from the top" and encourages retained lag with late release (Cochran and Stobbs 1968; T. P. Jorgensen 1999) — but a 2-DOF torque-program result is evidence about mechanism, not a prescription for any individual player, whose constraints (mobility, strength, consistency, injury history (Cole and Grimshaw 2016; Hume, Keogh, and Reid 2005)) sit outside this model. Any coaching translation should also respect that within-player timing variability in experts is real and partly functional (Neal et al. 2007; Langdown, Bridge, and Li 2012; Glazier 2011).

Chapter 9

Conclusions and Future Work

9.1 Conclusions

Proximal-to-distal sequencing in the golf swing is real, robust, and — as a coaching target — chronically over-interpreted. The mechanical literature, from Cochran and Stobbs through modern optimal control, does not support "pass the energy along as soon as possible"; it supports a subtler program: *build the energy proximally with the club held folded, defer the handoff, and let the transfer run — mostly parametrically, with a brief late assist — when the centrifugal field can pay for it.* Skilled players differ from amateurs along exactly this axis: not in whether their segments peak in order, but in how much energy the proximal system holds at transition and how late and how efficiently it is handed to the club.

This document adds the first committed, reproducible quantitative test of that proposition inside UpstreamDrift. On the platform's two-segment golf model, under matched proximal effort and a consistently defined first-pass impact: driving the wrist from the top cost 15.3% of clubhead speed relative to a fully passive wrist; the identical wrist torque applied at its optimal late onset gained 12.4%; and a program that actively restrained the wrist early (performing measurably negative early wrist work) and drove it late was best of all at both shoulder-torque levels. The energy accounting located the mechanism — early restraint banks arm energy that the passive joint-force channel converts to club energy late — and the pointwise counterfactual decomposition exhibited the predicted signature: **control acts early as retention; drift delivers late.** Within its model class, the retain-early/release-late hypothesis is not merely consistent with the evidence; it is what the model does when allowed to.

The claim is deliberately bounded: a planar 2-DOF chain with a fixed hub, rigid shaft, and unbounded constant torques demonstrates mechanism, not generality. The document's contribution is a *pathway*: hypotheses stated in falsifiable counterfactual terms, an instrumented model family of increasing fidelity, tested decomposition contracts, and reproducible experiment infrastructure that the remaining steps can reuse unchanged.

9.2 Future work

Tracked in issues #8377 and #8378, in dependency order:

1. **E3 — Re-simulation parity.** Implement the killswitch re-simulation ZTCF in Python on the backend-neutral model; compare pointwise and re-simulation profiles and the MATLAB pipeline on matched trajectories; document agreement and divergence.
2. **Bounded actuators and optimal control.** Replace constant torques with torque-velocity-limited profiles and re-run E1; then pose the onset question as an optimal-control problem and compare the optimizer's wrist-power time course against the S2 prediction.

3. **Model-fidelity ladder.** Repeat E1/E2/E4 on the triple pendulum (adding a torso link), the 8-DOF closed-loop golfer (two arms, shared club), and the MuJoCo humanoid, with cross-engine validation at each rung; add mobile-hub and flexible-shaft variants to expose the Miura/Nesbit and shaft mechanisms (Miura 2001; Nesbit and McGinnis 2014; MacKenzie and Sprigings 2009).
4. **Segmental energy-flow metrics in the shared layer.** Promote the experiment package's Robertson-Winter accounting into the engine-agnostic analysis library (full wrench-twist transfer, proximal braking efficiency), closing the platform gaps of
   4.
5. **E5 — Human-data replication.** Run the counterfactual and interface-power analyses on measured swings ingested through the platform's motion pipeline, across skill levels, testing the predictions of Section 4.5 observationally.
6. **Publication.** On completion, migrate this document to the affinedrift repository (issue #8379), converting repository-relative links to absolute URLs.

Acknowledgments

Prepared within the UpstreamDrift project for eventual publication in the affinedrift repository. The platform inventory reflects the codebase at the commit referenced by the pull request introducing this document; simulation data provenance (git SHA, parameters, integrator settings) is recorded in data/results_summary.json.

Appendix A: Platform Capability Inventory

Beyond the summary of Section 5, the counterfactual and energetics stack comprises: per-engine `compute_ztcf` / `compute_zvcf` / drift-control implementations (MuJoCo, Drake, Pinocchio, OpenSim, MyoSuite, analytic pendulum engines, Simscape bridge) behind the engine-core contract (`_dynamics_interface.py`); the pointwise evaluator with canonical-v2 trajectory input and HDF5 persistence (`ztcf_zvcf.py`); a Newton-Euler ZTCF force module with an analytic single-pendulum validation identity (`biomechanics/ztcf.py`); induced-acceleration analyzers (`induced_acceleration.py`); an orchestration layer with capability gating and REST/UI surfaces (`analysis/orchestrator.py`); MuJoCo inter-segment power flow with per-body energies and drift/control work decomposition (`power_flow.py`); engine-agnostic work/power/impulse and system-energy metrics (`power_work_metrics.py`, `energy_metrics.py`); conservation monitoring (`energy_monitor.py`); kinematic-sequence and coordination analytics with plot renderers (`kinematic_sequence.py`, `coordination_metrics.py`, `_coordination_sequence.py`); an approved internal research review of the kinematic-sequence metric (`kinematic_sequence.md`); and the MATLAB BASE/ZTCF/DELTA/ZVCF pipeline with per-joint linear/angular work and power decompositions (Section 4.4).

Known gaps

Table 9.1: Known platform gaps (tracked in issue #8378).

Gap	Consequence for this document
Two ZTCF operationalizations (pointwise vs re-simulation) coexist; no Python re-simulation ZTCF	Section 7 uses the pointwise variant only; parity untested (issue #8377)
Inter-segment power transfer in the shared layer is joint-power only (MuJoCo)	The Robertson-Winter accounting of Section 6 lives in the experiment package, pending promotion (issue #8378)
`ConservationMonitor` potential energy is first-order approximate	Not used for any number here; energies computed analytically
Flexible-shaft strain energy absent from the energy budget	Shaft recoil energy invisible until the shaft rung of Section 9.2
Proximal-braking-efficiency metric is an open TODO	Deceleration claims here rely on directly computed energies instead

Appendix B: Reproducibility Guide

From the repository root (Python 3.11+, with `numpy`, `matplotlib`, `pydantic`, `simpleeval`, `pandas` installed):

```
# 1. Run all experiments (E1 sweep, E2 counterfactuals, E4 powers)
python3 -m scripts.research.proximal_distal_energy.run_experiments

# 2. Bounded-actuator sensitivity sweep (E1b, incl. its figure)
python3 -m scripts.research.proximal_distal_energy.e1b_bounded_torque

# 3. Regenerate the remaining figures
python3 -m scripts.research.proximal_distal_energy.make_figures

# 4. Rebuild this document (Quarto + LaTeX)
cd docs/research/proximal_distal_energy_transfer
quarto render proximal_distal_energy_transfer.qmd --to pdf
```

Outputs land in `data/` (JSON + NPZ, with provenance: git SHA, integrator settings, full parameter values) and `figures/` (PDF). The experiments are deterministic (fixed-step RK4, no stochastic elements), so re-running reproduces every number in Section 7 exactly, up to floating-point environment differences.

Appendix C: Validation Evidence

The counterfactual stack this document relies on is contract-tested in CI: the pointwise evaluator reproduces the analytic double-pendulum ground truth to 10^{-9} via the ODE and analytical providers, and agrees with the independent MuJoCo backend to 10^{-7} when that engine is available (`test_ztcf_zvcf.py`); drift-plus-control reconstructs the measured acceleration (`test_counterfactual_experiments.py`); work-energy-theorem and conservation-law property tests run in CI (`tests/integration/conservation_laws/`); and a cross-engine comparison service produces divergence-annotated ZTCF/ZVCF panels between backends (`cross_engine_comparison.md`). The design guidelines make the drift-control decomposition and ZTCF/ZVCF availability normative across engines (`project_design_guidelines.qmd`). Within this document, the wrist-interface accounting carries its own internal check: the club's mechanical-energy rate matches the summed interface powers along the full trajectory (Figure 7.7, right).

Appendix D: Model Parameters and Program Definitions

Table 9.2: Model parameters and experiment constants (`scripts/research/proximal_distal_energy/`).

Parameter	Value	Notes
Arm length l_1	0.75 m	`GolfModelParams.default()`
Arm mass m_1	7.5 kg	both arms lumped
Arm COM ratio	0.45	$l_{c1} = 0.3375$ m
Arm inertia about COM	$0.3516\ \mathrm{kg \cdot m^2}$	about hub: $1.2059\ \mathrm{kg \cdot m^2}$
Club length l_2	1.0 m	
Shaft mass	0.15 kg	COM ratio 0.43
Clubhead mass	0.20 kg	point mass at tip
Club composite mass m_2	0.35 kg	$l_{c2} = 0.7557$ m
Club inertia about COM	$0.0403\ \mathrm{kg \cdot m^2}$	about wrist: $0.2402\ \mathrm{kg \cdot m^2}$
Plane inclination	35°	projected gravity $8.033\ \mathrm{m/s^2}$
Damping (shoulder, wrist)	$0.4, 0.25\ \mathrm{N \cdot m \cdot s/rad}$	viscous
Integrator	RK4, $\Delta t = 1$ ms	ODE reference backend
Initial state	$\theta_1 = -2.2$, $\theta_2 = -1.57$ rad, $\dot{q} = 0$	top of backswing
Shoulder torque τ_s	$60\ /\ 100\ \mathrm{N \cdot m}$	constant from $t = 0$
Wrist drive	$+15\ \mathrm{N \cdot m}$ from t_{on}	onset grid 0–0.35 s, step 0.025
Wrist restraint R	$5\ /\ 10\ \mathrm{N \cdot m}$ before t_{on}	restrain-then-drive only
Impact	first $\theta_1 + \theta_2 = 0$ crossing	valid iff $\theta_1 \leq 2.0$ rad

Data dictionary for the recorded outputs:

- `data/e1_sweep.json` — one row per program: profile, torques, onset, impact time, clubhead speed, arm angle at impact (null = invalid), plus a provenance block.
- `data/representative_traces.npz` — per representative program: `t, q, v, u`, drift/control splits, segment energies, clubhead-speed trace, and all interface-power series.
- `data/results_summary.json` — program definitions, impact tuples, phase budgets, passive baselines, best/worst rows, provenance.
- `data/e1b_bounded_sweep.json` — the bounded-actuator sweep rows with their own provenance block (isometric torques, ω_{max} values).

References

Anderson, Brady C., Ian C. Wright, and Darren J. Stefanyshyn. 2006. "Segmental Sequencing of Kinetic Energy in the Golf Swing." In *The Engineering of Sport 6*, 1:167–72. New York, NY: Springer. https://doi.org/10.1007/978-0-387-46050-5_30.

Balzerson, Daniel, Joydeep Banerjee, and John McPhee. 2016. "A Three-Dimensional Forward Dynamic Model of the Golf Swing Optimized for Ball Carry Distance." *Sports Engineering* 19: 237–50. https://doi.org/10.1007/s12283-016-0197-7.

Betzler, Nils F., Stuart A. Monk, Eric S. Wallace, and Steve R. Otto. 2012. "Effects of Golf Shaft Stiffness on Strain, Clubhead Presentation and Wrist Kinematics." *Sports Biomechanics* 11 (2): 223–38. https://doi.org/10.1080/14763141.2012.681796.

Bunn, John W. 1972. *Scientific Principles of Coaching*. 2nd ed. Englewood Cliffs, NJ: Prentice-Hall.

Cheetham, Phillip J., Gregory A. Rose, Richard N. Hinrichs, Robert J. Neal, Rob E. Mottram, Paul D. Hurrion, and Peter F. Vint. 2008. "Comparison of Kinematic Sequence Parameters Between Amateur and Professional Golfers." In *Science and Golf v: Proceedings of the World Scientific Congress of Golf*, edited by Debbie Crews and Rafer Lutz. Mesa, AZ: Energy in Motion. https://www.researchgate.net/publication/252236426.

Chu, Yungchien, Timothy C. Sell, and Scott M. Lephart. 2010. "The Relationship Between Biomechanical Variables and Driving Performance During the Golf Swing." *Journal of Sports Sciences* 28 (11): 1251–59. https://doi.org/10.1080/02640414.2010.507249.

Cochran, Alastair, and John Stobbs. 1968. *Search for the Perfect Swing*. Philadelphia: J. B. Lippincott.

Cole, Michael H., and Paul N. Grimshaw. 2016. "The Biomechanics of the Modern Golf Swing: Implications for Lower Back Injuries." *Sports Medicine* 46 (3): 339–51. https://doi.org/10.1007/s40279-015-0429-1.

Coleman, Simon, and Alan J. Rankin. 2005. "A Three-Dimensional Examination of the Planar Nature of the Golf Swing." *Journal of Sports Sciences* 23 (3): 227–34. https://doi.org/10.1080/02640410410001730179.

Dillman, Charles J., and Gregory W. Lange. 1994. "How Has Biomechanics Contributed to the Understanding of the Golf Swing?" In *Science and Golf II: Proceedings of the World Scientific Congress of Golf*, edited by Alastair J. Cochran and Martin R. Farrally. London: E & FN Spon. https://doi.org/10.4324/9780203474709-2.

Featherstone, Roy. 2008. *Rigid Body Dynamics Algorithms*. New York, NY: Springer. https://doi.org/10.1007/978-1-4899-7560-7.

Feltner, Michael E., and Jesús Dapena. 1986. "Dynamics of the Shoulder and Elbow Joints of the Throwing Arm During a Baseball Pitch." *International Journal of Sport Biomechanics* 2 (4): 235–59. https://journals.humankinetics.com/view/journals/jab/2/4/article-p235.xml.

"Foot–Ground Interaction and Clubhead Speed: Impulse-Based Energy Transfer as the Key Mechanism in the Golf Swing." 2026. *Frontiers in Sports and Active Living* 8. https://doi.org/10.3389/fspor.2026.1790645.

Fradkin, Andrea J., Caroline A. Sherman, and Caroline F. Finch. 2004. "How Well Does Club Head Speed Correlate with Golf Handicaps?" *Journal of Science and Medicine in Sport* 7 (4): 465–72. https://doi.org/10.1016/S1440-2440(04)80265-2.

Glazier, Paul S. 2011. "Movement Variability in the Golf Swing: Theoretical, Methodological, and Practical Issues." *Research Quarterly for Exercise and Sport* 82 (2): 157–61. https://doi.org/10.1080/02701367.2011.10599742.

Han, Ki Hoon, Christopher Como, Jemin Kim, Sangwoo Lee, Jaewoong Kim, Dae Kyoo Kim, and Young-

Hoo Kwon. 2019. "Effects of the Golfer–Ground Interaction on Clubhead Speed in Skilled Male Golfers." *Sports Biomechanics* 18 (2): 115–34. https://pubmed.ncbi.nlm.nih.gov/31042142/.

Hellström, John. 2009. "Competitive Elite Golf: A Review of the Relationships Between Playing Results, Technique and Physique." *Sports Medicine* 39 (9): 723–41. https://doi.org/10.2165/11315200-000000000-00000.

Herring, R. M., and A. E. Chapman. 1992. "Effects of Changes in Segmental Values and Timing of Both Torque and Torque Reversal in Simulated Throws." *Journal of Biomechanics* 25 (10): 1173–84. https://doi.org/10.1016/0021-9290(92)90073-A.

Hume, Patria A., Justin Keogh, and Duncan Reid. 2005. "The Role of Biomechanics in Maximising Distance and Accuracy of Golf Shots." *Sports Medicine* 35 (5): 429–49. https://doi.org/10.2165/00007256-200535050-00005.

Jorgensen, Theodore. 1970. "On the Dynamics of the Swing of a Golf Club." *American Journal of Physics* 38 (5): 644–51. https://doi.org/10.1119/1.1976419.

Jorgensen, Theodore P. 1999. *The Physics of Golf.* 2nd ed. New York: Springer / AIP Press.

Joyce, Christopher, Angus Burnett, Jodie Cochrane, and Kevin Ball. 2013. "Three-Dimensional Trunk Kinematics in Golf: Between-Club Differences and Relationships to Clubhead Speed." *Sports Biomechanics* 12 (2): 108–20. https://doi.org/10.1080/14763141.2012.728244.

Kenny, Ian C., Adam J. McCloy, Eric S. Wallace, and Steve R. Otto. 2008. "Segmental Sequencing of Kinetic Energy in a Computer-Simulated Golf Swing." *Sports Engineering* 11 (1): 37–45. https://doi.org/10.1007/s12283-008-0005-0.

Kwon, Young-Hoo, Christopher S. Como, Kunal Singhal, Sangwoo Lee, and Ki Hoon Han. 2012. "Assessment of Planarity of the Golf Swing Based on the Functional Swing Plane of the Clubhead and Motion Planes of the Body Points." *Sports Biomechanics* 11 (2): 127–48. https://doi.org/10.1080/14763141.2012.660799.

Kwon, Young-Hoo, Ki Hoon Han, Christopher S. Como, Sangwoo Lee, and Kunal Singhal. 2013. "Validity of the X-Factor Computation Methods and Relationship Between the X-Factor Parameters and Clubhead Velocity in Skilled Golfers." *Sports Biomechanics* 12 (3): 231–46. https://www.researchgate.net/publication/258639892.

Lampsa, Michael A. 1975. "Maximizing Distance of the Golf Drive: An Optimal Control Study." *Journal of Dynamic Systems, Measurement, and Control* 97 (4): 362–67. https://asmedigitalcollection.asme.org/dynamicsystems/article-abstract/97/4/362/400950.

Langdown, Ben L., Matthew Bridge, and François-Xavier Li. 2012. "Movement Variability in the Golf Swing." *Sports Biomechanics* 11 (2): 273–87. https://doi.org/10.1080/14763141.2011.650187.

Lee, Junghoon, Justin Ehrlich, and Christopher Cain. 2026. "The Effects of Different Kinematic Sequences of the Elite Golf Swing on Clubhead Speed." *International Journal of Sports Science & Coaching*. https://doi.org/10.1177/17479541261429479.

Lindsay, David M., Shannon Mantrop, and Anthony A. Vandervoort. 2008. "A Review of Biomechanical Differences Between Golfers of Varied Skill Levels." *International Journal of Sports Science & Coaching* 3 (Suppl. 1): 187–97. https://doi.org/10.1260/174795408785024117.

MacKenzie, Sasho J., and Eric J. Sprigings. 2009. "A Three-Dimensional Forward Dynamics Model of the Golf Swing." *Sports Engineering* 11 (4): 165–75. https://doi.org/10.1007/s12283-009-0020-9.

Madrid, Braden, Marco A. Avalos, Nicholas A. Levine, Noelle J. Tuttle, Kevin A. Becker, and Young-Hoo Kwon. 2020. "Association Between the on-Plane Angular Motions of the Axle-Chain System and Clubhead Speed in Skilled Male Golfers." *Applied Sciences* 10 (17): 5728. https://doi.org/10.3390/app10175728.

Marsan, Thibault, Maxime Bourgain, Patricia Thoreux, Christophe Sauret, and Philippe Rouch. 2019. "Biomechanical Analysis of the Golf Swing: Methodological Effect of Angular Velocity Component on the Identification of the Kinematic Sequence." *Acta of Bioengineering and Biomechanics* 21: 115–20. https://www.researchgate.net/publication/333377321.

Marshall, Robert N., and Bruce C. Elliott. 2000. "Long-Axis Rotation: The Missing Link in Proximal-to-Distal Segmental Sequencing." *Journal of Sports Sciences* 18 (4): 247–54. https://doi.org/10.1080/026404100364983.

McNally, William, and John McPhee. 2018. "Dynamic Optimization of the Golf Swing Using a Six Degree-of-Freedom Biomechanical Model." In *Proceedings*, 2:243. 6. https://doi.org/10.3390/proceedings2060243.

McPhee, John. 2022. "A Review of Dynamic Models and Measurements in Golf." *Sports Engineering* 25.

https://doi.org/10.1007/s12283-022-00387-0.

McTeigue, Michael, Steven R. Lamb, Robert Mottram, and Francis Pirozzolo. 1994. "Spine and Hip Motion Analysis During the Golf Swing." In *Science and Golf II: Proceedings of the World Scientific Congress of Golf*, edited by Alastair J. Cochran and Martin R. Farrally, 50–58. London: E & FN Spon. https://doi.org/10.4324/9780203474709-9.

Meister, David W., Amy L. Ladd, Erin E. Butler, Betty Zhao, Andrew P. Rogers, Conrad J. Ray, and Jessica Rose. 2011. "Rotational Biomechanics of the Elite Golf Swing: Benchmarks for Amateurs." *Journal of Applied Biomechanics* 27 (3): 242–51. https://doi.org/10.1123/jab.27.3.242.

Milburn, Peter D. 1982. "Summation of Segmental Velocities in the Golf Swing." *Medicine and Science in Sports and Exercise* 14 (1): 60–64. https://pubmed.ncbi.nlm.nih.gov/7070260/.

Miura, Ken. 2001. "Parametric Acceleration — the Effect of Inward Pull of the Golf Club at Impact Stage." *Sports Engineering* 4 (2): 75–86. https://doi.org/10.1046/j.1460-2687.2001.00071.x.

Myers, Joseph, Scott Lephart, Yung-Shen Tsai, Timothy Sell, James Smoliga, and John Jolly. 2008. "The Role of Upper Torso and Pelvis Rotation in Driving Performance During the Golf Swing." *Journal of Sports Sciences* 26 (2): 181–88. https://doi.org/10.1080/02640410701373543.

Neal, Robert J., Ryan Lumsden, Mark Holland, and Bruce Mason. 2007. "Body Segment Sequencing and Timing in Golf." *International Journal of Sports Science & Coaching* 2 (Suppl. 1). https://doi.org/10.1260/174795407789705497.

Nesbit, Steven M. 2005. "A Three Dimensional Kinematic and Kinetic Study of the Golf Swing." *Journal of Sports Science and Medicine* 4 (4): 499–519. https://pmc.ncbi.nlm.nih.gov/articles/PMC3899667/.

Nesbit, Steven M., and Ryan McGinnis. 2009. "Kinematic Analyses of the Golf Swing Hub Path and Its Role in Golfer/Club Kinetic Transfers." *Journal of Sports Science and Medicine* 8 (2): 235–46. https://www.jssm.org/volume08/iss2/cap/jssm-08-235.pdf.

Nesbit, Steven M., and Ryan S. McGinnis. 2014. "Kinetic Constrained Optimization of the Golf Swing Hub Path." *Journal of Sports Science and Medicine* 13 (4): 859–73. https://pmc.ncbi.nlm.nih.gov/articles/PMC4234956/.

Nesbit, Steven M., and Monika Serrano. 2005. "Work and Power Analysis of the Golf Swing." *Journal of Sports Science and Medicine* 4 (4): 520–33. https://www.jssm.org/volume04/iss4/cap/jssm-04-520.pdf.

Osis, Sean T., and Darren J. Stefanyshyn. 2012. "Golf Players Exhibit Changes to Grip Speed Parameters During Club Release in Response to Changes in Club Stiffness." *Human Movement Science* 31 (1): 91–100. https://doi.org/10.1016/j.humov.2011.02.006.

Penner, A. Raymond. 2003. "The Physics of Golf." *Reports on Progress in Physics* 66 (2): 131–71. https://doi.org/10.1088/0034-4885/66/2/202.

Pickering, W. M., and G. T. Vickers. 1999. "On the Double Pendulum Model of the Golf Swing." *Sports Engineering* 2 (3): 161–72. https://doi.org/10.1046/j.1460-2687.1999.00028.x.

Putnam, Carol A. 1991. "A Segment Interaction Analysis of Proximal-to-Distal Sequential Segment Motion Patterns." *Medicine and Science in Sports and Exercise* 23 (1): 130–44. https://pubmed.ncbi.nlm.nih.gov/1997807/.

———. 1993. "Sequential Motions of Body Segments in Striking and Throwing Skills: Descriptions and Explanations." *Journal of Biomechanics* 26 (Suppl. 1): 125–35. https://doi.org/10.1016/0021-9290(93)90084-R.

Robertson, D. Gordon E., and David A. Winter. 1980. "Mechanical Energy Generation, Absorption and Transfer Amongst Segments During Walking." *Journal of Biomechanics* 13 (10): 845–54. https://doi.org/10.1016/0021-9290(80)90172-4.

Sharp, Robin S. 2009. "On the Mechanics of the Golf Swing." *Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences* 465 (2102): 551–70. https://doi.org/10.1098/rspa.2008.0304.

Smith, Aimée, Jonathan Roberts, Eric S. Wallace, Pui Wah Kong, and Steph Forrester. 2015. "Golf Coaches' Perceptions of Key Technical Swing Parameters Compared to Biomechanical Literature." *International Journal of Sports Science & Coaching* 10 (4): 739–55. https://doi.org/10.1260/1747-9541.10.4.739.

Sprigings, Eric J., and Sasho J. MacKenzie. 2002. "Examining the Delayed Release in the Golf Swing Using Computer Simulation." *Sports Engineering* 5: 23–32. https://doi.org/10.1046/j.1460-2687.2002.00094.x.

Sprigings, Eric J., and Robert J. Neal. 2000. "An Insight into the Importance of Wrist Torque in Driving the Golfball: A Simulation Study." *Journal of Applied Biomechanics* 16 (4): 356–66. https://doi.org/10.1123/jab.16.4.356.

Tinmark, Fredrik, John Hellström, Kjartan Halvorsen, and Alf Thorstensson. 2010. "Elite Golfers' Kinematic Sequence in Full-Swing and Partial-Swing Shots." *Sports Biomechanics* 9 (4): 236–44. https://doi.org/10.1080/14763141.2010.535842.

White, Rod. 2006. "On the Efficiency of the Golf Swing." *American Journal of Physics* 74 (12): 1088–94. https://doi.org/10.1119/1.2346688.

Winter, David A. 2009. *Biomechanics and Motor Control of Human Movement.* 4th ed. Hoboken, NJ: John Wiley & Sons.

Zajac, Felix E., and Michael E. Gordon. 1989. "Determining Muscle's Force and Action in Multi-Articular Movement." *Exercise and Sport Sciences Reviews* 17: 187–230. https://pubmed.ncbi.nlm.nih.gov/2676547/.

Zajac, Felix E., Richard R. Neptune, and Steven A. Kautz. 2002. "Biomechanics and Muscle Coordination of Human Walking. Part I: Introduction to Concepts, Power Transfer, Dynamics and Simulations." *Gait & Posture* 16 (3): 215–32. https://doi.org/10.1016/S0966-6362(02)00068-1.

Zheng, Naiquan, Steve W. Barrentine, Glenn S. Fleisig, and James R. Andrews. 2008. "Kinematic Analysis of Swing in Pro and Amateur Golfers." *International Journal of Sports Medicine* 29 (6): 487–93. https://doi.org/10.1055/s-2008-1038732.